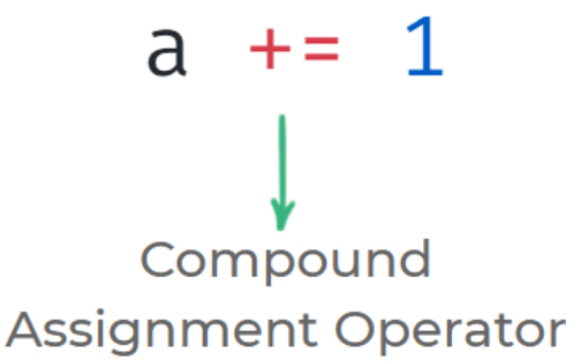
Problem Solving Session

Compound Assignment Operators



`a += 1` is similar to `a = a + 1`

Different compound assignment operators are `+=` , `-=` , `*=` , `/=` , `%=`

Code

PYTHON

```python
a = 1
a += 2
print(a)
```

Output

```
3
```

Code

PYTHON

```python
a = 10
a -= 2
print(a)
```

Output

```
8
```

Examples of Compound Assignment Operators

Code

PYTHON

```python
a = 10
a /= 2
print(a)
```

Output

```
5.0
```

Code

PYTHON

```python
a = 10
a %= 2
print(a)
```

Output

```
0
```

You will learn more detailed about compound assignment operator in further sessions.

Question 1 of 5

Which of the following is similar to a += 1 in Python?

- a = a + 1
- a = 1
- a = a * 1
- a = a - 1

Show Answer | Submit

Questions

Numbers in Hollow Square

Question:

Write a program to print a Hollow Square of N rows and N columns using numbers, where integer N is given as an input

```
1 2 3 4 5
1       5
1       5
1       5
1 2 3 4 5
```

Solution

PYTHON

```python
n = int(input())
for row_number in range(1, n + 1):
    for col_number in range(1, n + 1):
        if row_number == 1 or row_number == n or col_number == 1 or col_number =
            print(col_number, end=" ")
        else:
            print(" ", end=" ")
    print()
```

Question:

Numbers in Hollow Pyramid

Write a program to print a Hollow Pyramid of N rows using numbers starting from 5, where integer N is given as an input

```
    5
   5 6
  5   7
 5     8
5       9
5 6 7 8 9 10
```

Solution

PYTHON

```python
n = int(input())

for i in range(1, n + 1):
    if i == 1:
        row = ""
        row += (n - 1) * (" ") + "5 "
        print(row)
    elif (i==n):
        for j in range(n):
            print(5 + j, end=" ")
        print()
```

Expand

In the above solution, the compound assignment operator is used,

`row += (n - 1) * (" ") + "5 "` is similar to `row = row + (n - 1) * (" ") + "5 "`

Similarly,
`row += left_spaces * " " + "5 " + middle_spaces * " " + str(i + 4) + " "` is similar to
`row = row + left_spaces * " " + "5 " + middle_spaces * " " + str(i + 4) + " "`

Numbers in Inverted Hollow Triangle

Question:

Write a program to print a Hollow Right Angled Triangle of N rows using numbers starting from 1, where integer N is given as input

```
1
2 3
4   5
6 7 8 9
```

PYTHON

```python
n = int(input())

num = 1

for i in range(1, n + 1):
    for j in range(1, i + 1):
        if (i == n or i == j or j == 1):
            print(num, end = " ")
            num += 1
        else:
```

Expand

In the above solution, the compound assignment operator is used,

`num += 1` is similar to `num = num + 1`

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Notes

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